Experiment 10: Line Plot and Bar Plot Using Matplotlib

Aim

To create line and bar plots to visualize monthly sales data using Matplotlib.

Algorithm

1. Import Matplotlib.
2. Get month names and sales values as input.
3. Create a line plot using plt.plot().
4. Create a bar plot using plt.bar().
5. Display both plots.

Code

Experiment 10 Code

```
import matplotlib.pyplot as plt

months = input("Enter month names: ").split()
sales = list(map(int, input("Enter sales values: ").split()))

plt.plot(months, sales, marker='o')
plt.title("Monthly Sales - Line Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()

plt.bar(months, sales)
plt.title("Monthly Sales - Bar Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```

Input

Enter the following in Python IDLE when prompted:

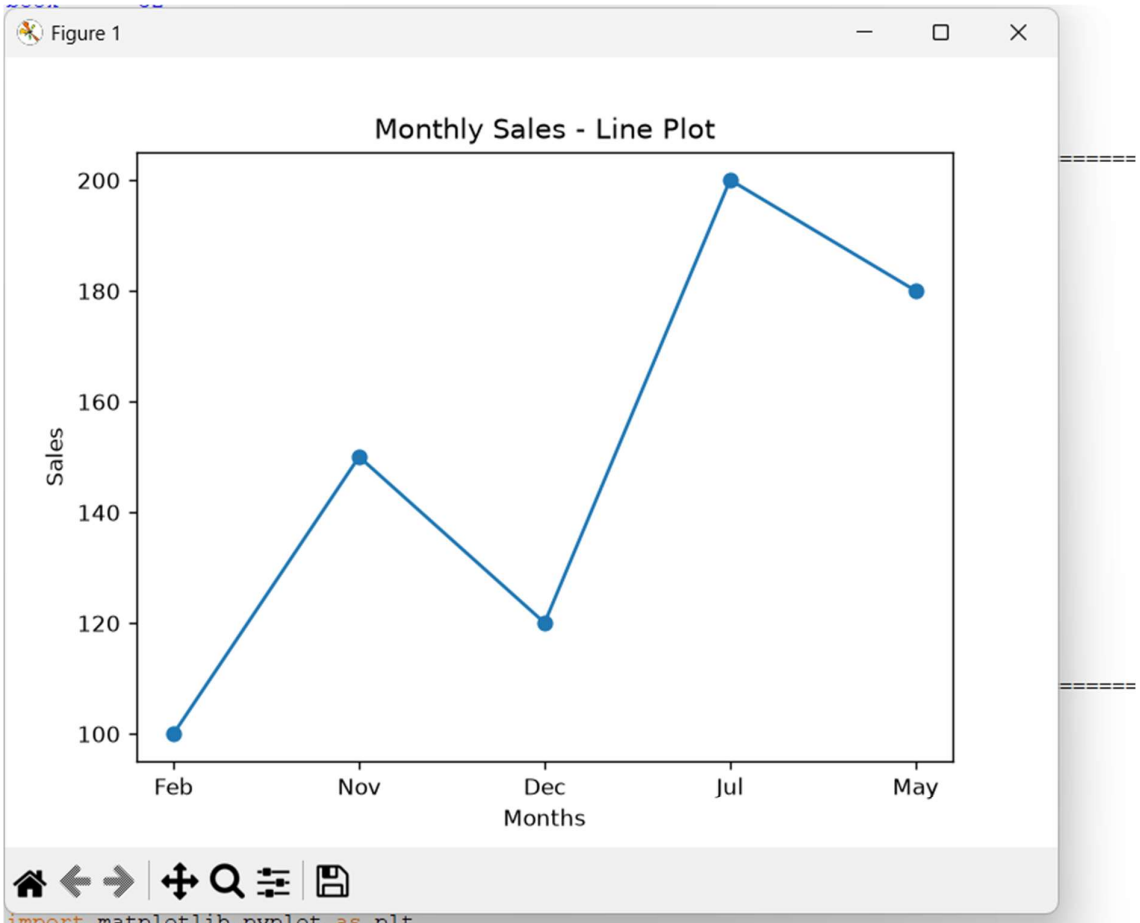
Enter month names: Jan Feb Mar Apr May

Enter sales values: 100 150 120 200 180

Output

- **Line plot:** Displays the monthly sales trend using points connected by lines.
- **Bar plot:** Displays monthly sales using vertical bars.

OUTPUT




```
import matplotlib.pyplot as plt

================ RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 10.py ============
Enter month names: Feb Nov Dec Jul May
Enter sales values: 100 150 120 200 180
```

Result

Thus, line and bar plots were successfully created to visualize monthly sales data using Matplotlib.